8. Automatic Cooling Bed with Twin Channel & Cold Shear
Size 66 Mtr.

a. Twin Channel

Application : Delivery of Rolled and cut Bars on to the Cooling Bed
Type : Electric Motor Driven single shaft Twin Channel
Bar Size : Dia 8mm – 32mm
Length : 66 Mtrs.
Maximum Speed : 20 M/Sec
Channels : S.G Iron Casting 1.1m long each
Central Box : 2.2m long Cast Iron, Air Cooled
Shafts : 2.2m Meter Long steel shaft
Bearings : Supported by pillow block
Cam : Forged Steel
Cam Followers: Roller Bearing Mounted on Pins in the C Channel
Coupling : Flexible
Drive : DC Electric Motor for line shaft operation

The Twin Channel bar delivery equipment consists of the following

- C type S.G. Iron Channels on both sides butting against cast Iron Central Box and hinged on top of the Box
- Box suspended through lugs from longitudinal frame on top
- Cam follower rollers mounted on a lug on each channel
- Single shaft is mounted on bearings and suspended from the longitudinal frame
- Cams for each C Channel are mounted on shaft
- Electric motor drive to main single shaft for movement

Operation

Both the C Channel (Right & Left) are closed. Rolled Re bar enters one of the C Channel and is cut to Cooling bed length while travelling by dividing shear. After cut, the bar tail end is braked using the tail braking pinch roll, so that the bar stops before its front end can shoot out of the channel. Motor operated cam shaft opens the C Channel to drop the bar on the Cooling bed straightening pockets. After dropping the bar, the channel closes immediately through rotation and is ready for the next bar. As soon the bar is cut, the next part of the bar is directed toward the other channel, which is ready to receive the same.

b. AUTOMATIC RAKE TYPE COOLING BED

Type	: Rake Type
Size	: 66 m long
Bar speed	: 22 m/sec (max)
In-let temperature of bar	: 600°C
Out-let temperature of bar	: 250°C

The cooling bed proposed is of automatic rake type and has the following:-
i) Fixed Grid
ii) Moving rakes
iii) Eccentrics and drives.
iv) Fixed rakes
v) Aligning roller

i) FIXED GRID

Grid length	: 66 m
Notch pitch	: 66 mm
Grid material	: cast iron

The fixed grid is a simple cast iron construction with notches profiled identical to rake notches. They receive bars from diverter aprons. The moving rakes picks up the material from the fixed grid. Recess is provided for the movement of rakes. The fixed grid is provided in suitably sized segments.

ii) MOVING RAKES

Rake Length	: To fit size
Rake Pitch	: 600 mm
Notch Pitch	: 66 mm
Rake Material	: MS flats
Rake Mounting	: On MS fabricated frame
Rake Material	: MS flats
Rake Mounting	: On MS fabricated frame
Rake Thickness	: 16 mm

The moving rakes transports the bars towards the discharge end moving the material by one notch for every cycle, during the period cooling the bar by natural currents. The movement of the rakes is through the eccentrics.

iii) ECCENTRICS & DRIVES

Rake Drive : Through eccentrics
Eccentricity : 36 mm
Line Shaft Bearings : Spherical roller bearings housed in standard Plummer Blocks
Line Shaft Material: EN8
Eccentric Material : EN9
Line Shaft Inter Couplings: Muff type

Couplings between Line shaft and Worm Reduction Gearbox: Half Gear Coupling No. 108

Couplings between Worm Reduction Gearboxes & Reduction Gearbox and Motor: Half Gear Coupling No. 105

Drive mechanical : Worm Gear Boxes
The eccentrics are mounted on line shafts, housed in bearing housing bolted to moving rake mounting frame. The line shafts are coupled to gearboxes driven by a DC motor. The arrangement provides for one rotation of eccentric per cycle, which in effect moves the material by bone rake pitch.

iv) FIXED RAKES

Rake Length : To fit size
Rake Pitch : 300 mm
Notch Pitch : 66 mm
Rake Material : MS flats
Rake Mounting : On beams anchored to foundation.
The fixed rakes hold the bars during the rest of the periods.

v) ALIGNMENT ROLLS

Roll Diameter : ~200 mm
Bearings : Anti friction
Roll Lengths : ~ 600 mm
Notch Profile : To suit rake
Roll Drive : Geared motor

Towards the discharge end are placed a series of aligning rolls which aligns the bar front ends to facilitate cutting at cold shear. These are cast iron driven rolls profiled identical to the rakes and positioned between rakes.

c. Run Out Roller Table **1 Set**

Run- out roller table
a) Roller Pitch – 1500mm
b) Roller size 216 Ø x 500 B.L.
c) Spherical roller bearing
d) MS Fabrication Table Frame
e) Length 66 Mtr.

d. Cold Shear **1 No.**

Capacity	: 300 Ton
Blade width	: 600 mm
Nos. of cuts	: 8 cuts/ min
Construction	: Fabricated steel body, heat-treated steel gears, shafts & blades, shafts mounted on roller bearings
Drives	: AC Motor

9 Continuous operating dividing shear for TMT Bars

Type : Direct Driven Dividing Shear (Steel Fabricated)
Bar Speed : Max. 20 m/s
High speeds Continuous Shear with Lubrication Piping, Oil Supply System, Servo Bar Shifter
Delivery Guide Trough & Valve Stand
Gears : Helical, Case Hardened & Ground with Surface Hardness HRC 58-60
Shafts : Heat Treated Steel
Blades : Heat Treated Steel (46-50 HRC)